

The main upshot of being able to find the maximum and minimum of a function is that if we can model a situation as a function, we can now find an optimal outcome. For example, if we have a function $C(x)$ that gives cost of production, we know how to find the point of minimum cost.

These problems are called 'optimization problems'. There are a lot of problems in which the solution involves optimizing some function; often the most difficult parts are recognizing that the problem requires optimization and what function is relevant.

Recall:

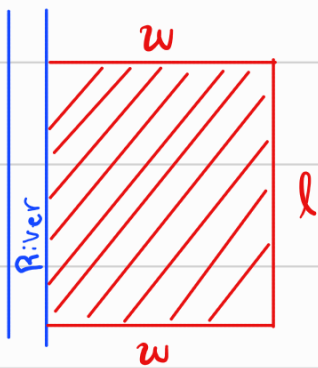
The closed interval Method

To find the maximum and minimum of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f for all critical numbers in (a, b)
2. Find the value of f at the end points a and b .
3. The largest and smallest value in 1 and 2 are the maximum and minimum respectively

Ex: A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a river. No fence is needed along the river. What are the dimensions of the field with maximum area?

Picture



Formulas

Area, $A = l \cdot w$

Fence: $\underline{2400 = l + 2w}$ $\sim ?$ No fence on river

l_{given}

Finding Function

Notice, $A = lw$ is a function of two variables.

this is bad as we can't use our calculus tricks.

However, we know $2400 = l + 2w$. So,

$$l = 2400 - 2w$$

$$A = (2400 - 2w)w = 2400w - 2w^2$$

And this is a function we can maximize like usual.

Finding Max

$$A(w) = 2400w - 2w^2$$

$$A'(w) = 2400 - 4w$$

$$A'(w) = 0$$

$$2400 - 4w = 0$$

$$\underline{600 = w}$$

$$\text{If } w = 600, \quad \underline{l = 2400 - 2w = 1200}$$

Total area is then $A(600) = 720,000$

Max Area is $720,000 \text{ ft}^2$

when $l = 1200 \text{ ft}$

$w = 600 \text{ ft}$

Ex: Find numbers $x, y > 0$ that sum to 8 but $x^2 + y^2$ is maximized.

Formulas

$$x + y = 8 \rightarrow y = 8 - x$$

Equation

$$\begin{aligned} f(x) &= x^2 + (8-x)^2 \\ &= 2x^2 - 16x + 64 \end{aligned}$$

Finding Max

$$f'(x) = 4x - 16$$

$$f'(x) = 0 \text{ if } x = 4$$

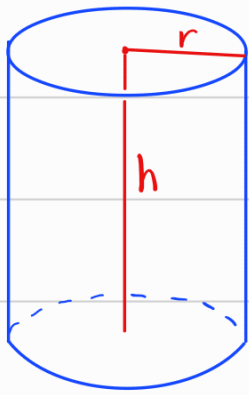
$$y = 8 - x = 8 - 4 = 4$$

$$x = 4$$

$$y = 4$$

Ex: A cylindrical can is made to hold 1 litre of liquid. Find the dimensions that will minimize the cost of the metal to manufacture the can.

↳ minimize material



$$V = 1L = \pi r^2 \cdot h$$

↳ 1000 cm³

$$\text{Surface Area} = 2\pi r^2 + 2\pi rh$$

↳ top and bottom circles ↳ rectangle wrapped around

We want to minimize surface area. However, it's currently has two variables.

$$1000 \text{ cm}^3 = \pi r^2 h, \text{ we have } h = \frac{1000}{\pi r^2}$$

$$\begin{aligned} \text{then } A &= 2\pi r^2 + 2\pi r \left(\frac{1000}{\pi r^2} \right) \\ &= 2\pi r^2 + \frac{2000}{r} \end{aligned}$$

To minimize

$$\begin{aligned} A'(r) &= 4\pi r - \frac{2000}{r^2} \\ &= \frac{4\pi r^3 - 2000}{r^2} \end{aligned}$$

$$A'(r) \text{ DNE if } r=0$$

$$A'(r)=0 \quad \text{if} \quad 4\pi r^3 - 2000 = 0$$

$$4\pi r^3 = 2000$$

$$r = \sqrt[3]{\frac{500}{\pi}}$$

$r=0$ doesn't make sense, We need to Check $r = \sqrt[3]{\frac{500}{\pi}}$ is a minimum. It's not hard to see $A'(r) < 0$ if $r < \sqrt[3]{\frac{500}{\pi}}$ and $A'(r) > 0$ if $r > \sqrt[3]{\frac{500}{\pi}}$. So $r = \sqrt[3]{\frac{500}{\pi}}$ cm is a min.

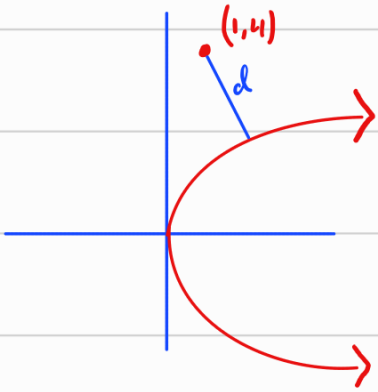
$$\begin{aligned} \text{If } r = \sqrt[3]{\frac{500}{\pi}}, \quad h &= \frac{1000}{\pi r^2} = \frac{1000}{\pi \left(\sqrt[3]{\frac{500}{\pi}}\right)^2} = \frac{2(500)}{\pi (500/\pi)^{2/3}} = 2 \left(\frac{500^{1/3}}{\pi^{1/3}} \right) \\ &= 2 \sqrt[3]{\frac{500}{\pi}} = 2r \end{aligned}$$

$$\begin{aligned} r &= \sqrt[3]{\frac{500}{\pi}} \\ h &= 2 \sqrt[3]{\frac{500}{\pi}} \end{aligned}$$

Ex: Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1, 4)$.

$$y^2 = 2x \rightarrow x = \frac{y^2}{2}$$

$$d = \sqrt{(x-1)^2 + (y-4)^2}$$



Since d is positive, d and d^2 attain a minimum at the same point.

So we can minimize d^2 instead, which is easier

$$\begin{aligned} f(y) = d^2 &= (x-1)^2 + (y-4)^2 \\ &= \left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2 \\ &= \frac{y^4}{4} - y^2 + 1 + y^2 - 8y + 16 \\ &= \frac{y^4}{4} - 8y + 17 \end{aligned}$$

$$f'(y) = y^3 - 8$$

$$f'(y) = 0 \text{ if } y = 2$$

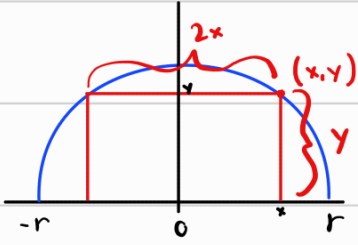
We should check this is a min.

$f'(y) < 0$ if $y < 2$ and $f'(y) > 0$ if $y > 2$.

$$\text{If } y = 2, \quad x = \frac{y^2}{2} = \frac{4}{2} = 2$$

So the point $(2, 2)$ on $y^2 = 2x$ is closest to $(1, 4)$.
($d = \sqrt{5}$).

Ex: Find the area of the largest rectangle that can be inscribed in a semicircle of radius r .



$$A = 2xy$$

$$\text{Circle: } x^2 + y^2 = r^2 \quad \text{✓ } r \text{ is fixed}$$

$$\hookrightarrow y = \sqrt{r^2 - x^2}$$

$$A = 2x\sqrt{r^2 - x^2}$$

$$A' = 2\sqrt{r^2 - x^2} + 2x \cdot \frac{1}{2} \frac{-2x}{\sqrt{r^2 - x^2}}$$

$$= \frac{2(r^2 - x^2)}{\sqrt{r^2 - x^2}} - \frac{2x^2}{\sqrt{r^2 - x^2}}$$

$$= \frac{2r^2 - 4x^2}{\sqrt{r^2 - x^2}}$$

A' DNE if $x=r$, but this doesn't make sense in the problem.

$$A' = 0 \quad \text{if} \quad 2r^2 - 4x^2 = 0$$

$$r^2 = 2x^2$$

$$\underline{\frac{r}{\sqrt{2}} = x}$$

$$A\left(\frac{r}{\sqrt{2}}\right) = 2 \frac{r}{\sqrt{2}} \sqrt{r^2 - \frac{r^2}{2}} = 2 \frac{r}{\sqrt{2}} \sqrt{\frac{r^2}{2}} = 2 \frac{r^2}{2} = r^2$$

Maximum area is r^2